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Erratum: gap in the proof of EnKF convergence

Concerns

- M. D. Butala, J. Yun, Y. Chen, R. A. Frazin, F. Kamalabadi, “Asymptotic convergence of the ensemble Kalman filter,” *ICIP 2008*, pp. 825–828. (thesis ref. [121])
- M. D. Butala, R. A. Frazin, Y. Chen, F. Kamalabadi, “Tomographic imaging of dynamic objects with the ensemble Kalman filter,” *IEEE TIP* **18**(8):1573–1587, 2009. (thesis ref. [6] in Mandel et al.)
- M. D. Butala, PhD thesis, App. A (Convergence Proof), proof of Thm 4.1.

Claimed by J. Mandel, L. Cobb, J. D. Beezley, “On the convergence of the ensemble Kalman filter,” *Appl. Math.* **56**(6):533–541, 2011 (arXiv:0901.2951v2, p. 2):

“The proof of EnKF convergence in [6] has a gap; it assumes that certain covariances derived from the ensemble exist, which is not guaranteed without an L2 bound.”

Verdict. The gap is real and Mandel’s description is accurate. It is localized to the cross-term argument in Step 2 of App. A, eqs. (A.13)–(A.18) (and the analogous step in (A.22)). It is repaired by inserting one a priori moment-bound lemma (below). The theorem statement survives unchanged; the proof can in fact be strengthened from convergence in probability to L^p convergence.

1. Notation (thesis App. A)

Ensemble size L , state dim N , time index i . Members $\hat{x}_{i|i-1}^l$, sample mean $\hat{x}_{i|i-1}$, sample covariance $\hat{P}_{i|i-1}^f$ as in (4.17)–(4.20). Taper C_i , gain $\hat{K}_i$ as in (4.19). Perturbed measurements $y_i^l = y_i + v_i^l$ with $v_i^l \stackrel{\text{iid}}{\sim} N(0, R_i)$; process noise $u_i^l \stackrel{\text{iid}}{\sim} N(0, Q_i)$. Anomalies $z_{i|i-1}^l = \hat{x}_{i|i-1}^l - \hat{x}_{i|i-1}$, so $\sum_{l=1}^L z_{i|i-1}^l = 0$.

Induction hypotheses used in Step 2:

$$\hat{x}_{i|i-1} \xrightarrow{p} x_{i|i-1}^\infty, \quad \hat{P}_{i|i-1}^f \xrightarrow{p} P_{i|i-1}^\infty. \quad (\text{A.4–A.5})$$

2. The gap

The posterior sample covariance (A.10) contains the cross terms

$$\frac{1}{L-1} \sum_{l=1}^L z_{i|i-1}^l (w_i^l)^T, \quad w_i^l := y_i^l - \bar{y}_i, \quad (\text{A.12})$$

which must be shown to converge in probability to 0. Componentwise, (A.13)–(A.15) reduce this to the average of

$$W_{L,l} := z_{i|i-1,m}^l v_{i,n}^l.$$

Two steps of the argument presuppose that these ensemble-derived random variables have finite, L -uniform second moments:

1. **(A.17)** asserts $\text{Cov}(z_{i|i-1,m}^k v_{i,n}^k, z_{i|i-1,m}^l v_{i,n}^l) = 0$ for $k \neq l$. A *covariance* is being computed; it exists only if $W_{L,k} W_{L,l}$ and $W_{L,l}^2$ are integrable. This is exactly Mandel’s “certain covariances derived from the ensemble.”
2. **(A.18)** invokes the L^2 weak law of large numbers (Durrett, *Probability: Theory and Examples*, 2nd ed., p. 36) on $\sum_l W_{L,l}$. That result requires $\text{Var}(W_{L,l}) \leq C$ **uniformly in L** , so that $\text{Var}((L-1)^{-1} \sum_l W_{L,l}) \leq C/(L-1) \rightarrow 0$.

Neither is established anywhere. The induction hypotheses (A.4)–(A.5) assert only *convergence in probability* of the sample mean and sample covariance. That places no constraint on the moments of the members: $\hat{P}_{i|i-1}^f$ is a.s. finite for every finite L under no moment assumption at all, and (A.4)–(A.5) are compatible with $\sup_L \mathbb{E}[(z_{i|i-1,m}^L)^2] = \infty$. In that case $\mathbb{E}[W_{L,l}^2] = \mathbb{E}[(z_m^L)^2] \mathbb{E}[(v_n^L)^2]$ can be infinite or grow with L , (A.17) is meaningless, the L^2 WLLN does not apply, and (A.12) is not shown to vanish.

The same omission propagates to the cross terms of (A.22) in Step 3 (“by a similar argument used to show that (A.12) converges in probability to the matrix 0”).

Secondary defect (not the gap). (A.16) cites “the WLLN” for $(L-1)^{-1} \sum_l z_{i|i-1,m}^l [y_i]_n \rightarrow 0$. The summands are *not* i.i.d. (each z^l contains the shared sample mean $\hat{x}_{i|i-1}$), so the i.i.d. WLLN does not apply as cited, and the displayed LHS of (A.16) writes $[y_i]_n$ where the argument requires $[y_i]_n$. The conclusion is nonetheless correct and needs no LLN: $\sum_l z_{i|i-1}^l = 0$ implies $\sum_l z_{i|i-1,m}^l [y_i]_n = 0$ *identically*.

3. Repair

Lemma 1 (a priori L^p bound; = Mandel–Cobb–Beezley Lemma 5, extended to tapers)

For every time index i and every $p \in [1, \infty)$ there is a constant $c(i, p) < \infty$, independent of L , such that for all $L \geq 2$ and all l

$$\|\hat{x}_{i|i-1}^l\|_p, \|\hat{x}_{i|i}^l\|_p, \|\hat{P}_{i|i-1}^f\|_p, \|\hat{K}_i\|_p \leq c(i, p).$$

Proof. Induction on i . The initial ensemble (4.18) is Gaussian, so all moments exist with bounds independent of L . Assume the bound at $i-1$.

- *Forecast (4.22):* $\|\hat{x}_{i|i-1}^l\|_p = \|F_{i-1} \hat{x}_{i-1|i-1}^l + u_{i-1}^l\|_p \leq \|F_{i-1}\| c(i-1, p) + \|u\|_p$.

- *Mean:* $\|\hat{x}_{i|i-1}\|_p \leq L^{-1} \sum_l \|\hat{x}_{i|i-1}^l\|_p \leq c$ by Jensen.
- *Anomalies and sample covariance:* $\|z^l\|_p \leq 2c$, and by the Cauchy–Schwarz bound $\|WZ\|_p \leq \|W\|_{2p} \|Z\|_{2p}$ (Mandel (4.1)), $\|\hat{P}_{i|i-1}^f\|_p \leq (L-1)^{-1} \sum_l \|z^l (z^l)^T\|_p \leq \|z^1\|_{2p}^2 \leq c'$.
- *Taper:* $P \mapsto C_i \circ P$ is linear and bounded in any matrix norm (tapers obey $|[C_i]_{jk}| \leq 1$; more generally $\|C \circ P\| \leq \|C\|_{\max} \|P\|$), so $\|C_i \circ \hat{P}_{i|i-1}^f\|_p \leq c''$.
- *Gain:* $H_i(C_i \circ \hat{P}^f)H_i^T + R_i \succeq R_i \succ 0$, hence $\|(H_i(C_i \circ \hat{P}^f)H_i^T + R_i)^{-1}\| \leq \|R_i^{-1}\|$ **deterministically** (no inverse moments are needed), and $\|\hat{K}_i\|_p \leq c'' \|H_i^T\| \|R_i^{-1}\|$.
- *Analysis (4.21):* with $y_i^l = y_i + v_i^l$, $\|\hat{x}_{i|i}^l\|_p \leq \|\hat{x}_{i|i-1}^l\|_p + \|\hat{K}_i y_i^l\|_p + \|\hat{K}_i H_i \hat{x}_{i|i-1}^l\|_p \leq c(i, p)$ by (4.1) and the induction hypothesis (which supplies every $2p$ as well). $\square$

Note the two load-bearing facts: $R_i \succ 0$ makes the random inverse uniformly bounded, and the hypothesis “for all p at once” absorbs the $p \mapsto 2p$ losses of Cauchy–Schwarz.

Lemma 2 (the missing step)

Fix i and let $W_{L,l} = z_{i|i-1,m}^l v_{i,n}^l$. Then $W_{L,l} \in L^2$ with $\sup_{L,l} \mathbb{E}[W_{L,l}^2] \leq C(i) < \infty$, and $\{W_{L,l}\}_{l=1}^L$ is pairwise uncorrelated with mean zero. Consequently $(L-1)^{-1} \sum_l W_{L,l} \rightarrow 0$ in L^2 and in probability.

Proof. $z_{i|i-1}^l$ is measurable w.r.t. the σ -algebra generated by the forecast ensemble, and v_i^l w.r.t. that generated by $\{v_i^k\}_k$; these are independent (the initial ensemble, $\{u^l\}$, $\{v^l\}$ are mutually independent). Hence by Lemma 1

$$\mathbb{E}[W_{L,l}^2] = \mathbb{E}[(z_{i|i-1,m}^l)^2] \mathbb{E}[(v_{i,n}^l)^2] \leq c(i, 2)^2 [R_i]_{nn} =: C(i),$$

which is the integrability that (A.17) requires. For $k \neq l$, $\mathbb{E}[W_{L,k} W_{L,l}] = \mathbb{E}[z^k z^l] \mathbb{E}[v^k v^l] = 0$ since the v^l are i.i.d. mean zero, and $\mathbb{E}[W_{L,l}] = 0$. The L^2 WLLN (Durrett, p. 36) now applies: $\text{Var}((L-1)^{-1} \sum_l W_{L,l}) \leq C(i)/(L-1) \rightarrow 0$. $\square$

Completion of the proof of Theorem 4.1

- **Step 1** is unchanged: (4.18) is Gaussian, so the i.i.d. WLLN applies to the sample mean and (entrywise, via $x_j x_k \in L^1$) to the sample covariance.
- **Step 2**, eqs. (A.12)–(A.18): replace (A.17)–(A.18) by Lemma 2. The cross terms (A.12) then converge in probability to 0; the first term of (A.10) converges by (A.5) and continuity of the Hadamard product; the fourth by the continuous mapping theorem and $R_i \succ 0$. Eq. (A.19) follows. For (A.16), delete the WLLN citation and use $\sum_l z_{i|i-1}^l = 0$ as noted in §2.
- **Step 3**, eq. (A.22): the cross terms in $\sum_l z_{i|i}^l (u_i^l - \bar{u}_i)^T$ and its transpose vanish by the *same* argument as Lemma 2 ($z_{i|i}^l \perp u_i^l$, $\sup_L \|z_{i|i}^l\|_2 < \infty$ by Lemma 1). The remaining terms converge by (A.19) and the i.i.d. WLLN on $\{u_i^l\}$. Eqs. (A.21), (A.23) follow, closing the induction.

Strengthening

Lemma 1 gives uniform L^p bounds on the members, hence uniform integrability. As in Mandel–Cobb–Beezley Thm 1, convergence in probability may then be upgraded to

$$\hat{x}_{i|i}^l \rightarrow x_{i|i}^\infty \quad \text{in } L^q \text{ for every } q < p, \quad \hat{x}_{i|i}, \hat{P}_{i|i}^f \rightarrow x_{i|i}^\infty, P_{i|i}^\infty \text{ in } L^q,$$

at each fixed time index i . Corollary 4.2 (unlocalised EnKF $\rightarrow$ KF/LMMSE) is unaffected.

4. Remarks

1. **Scope of Mandel–Cobb–Beezley.** Their paper treats only the *unlocalised stochastic* EnKF (Burgers–van Leeuwen–Evensen); they state explicitly that square-root variants “are not the subject of this paper.” Theorem 4.1’s tapered limit (EnKF $\rightarrow$ LKF, and hence the quantification of *localization bias* via $\text{KF} \neq \text{LKF}$) is outside their scope. The repair above is written for the tapered case and covers that content.
2. **Nature of the gap.** It is a missing hypothesis, not a false conclusion. No step of the argument is wrong once L -uniform second moments are available; the fix is a one-lemma insertion, not a restructuring.
3. **Horizon.** Theorem 4.1 is per fixed time index i (finite horizon, induction on i). Time-uniform / infinite-horizon accuracy is a separate question; see Cheng, Sanz-Alonso & Wan-iorek, arXiv:2609.23927 (2026) for deterministic square-root EnKFs, which also separates sampling error from localization bias.